

COVER LETTER - FRONTIER AI

JULY 28, 2026

Dear Hiring Team,

I am most useful where the problem matters, the path is unclear, and reliable progress must be built from first principles. Frontier roles differ in title, but they repeatedly need the same kind of person: someone who can reconstruct an unfamiliar domain, discover the missing decomposition, turn ambiguity into experiments and systems, and own the result end to end. The chronology below demonstrates that operating pattern:

- **Early life** — Raised by my grandparents on a subsistence farm in rural central China, I was working the fields by four. The farmer is humanity's oldest full-stack engineer — seed, soil, water, weather, tools, repair, timing, storage, one person owning all of it, with the harvest as the only test that counts. I was a full-stack engineer before I knew the word for it, and responsibility arrived before credentials. Every domain since has been entered the same way: from zero, whole stack, outcome owned.
- **High school** — At fifteen, I led a fifty-person robotics club. I negotiated directly with my principal to spend two of three years away from campus—the first such arrangement in the school's history—yet placed in my class's top five on China's national college entrance examination.
- **Age sixteen** — I entered Professor Zexiang Li's DJI, RoboMaster, and XbotPark ecosystem—a de facto “DJI Innovation High School”—and became final captain of RoboMaster's only high-school team.
- **France** — I taught myself French in six months and moved there, then spent three years in pure mathematics—deliberately at the structural layer: category and topos theory, higher categories, stable homotopy—the mathematics in which other mathematics is organized. Learning that layer once is why unfamiliar formal domains open quickly to me. At Orsay, I audited SGA and Weil-conjectures seminars and worked through SGA4/EGA. That search became [The Mathematics Ontology](#)  (*inspect the evidence*).
- **United States** — I arrived with almost no money, survived, and rebuilt a life. I learned patent law from thousand-page casebooks and carried Ergology from invention through patent, legal structure, partners, and production-ready documentation.
- **2026** — Full-stack, cold start, high agency — those are claims, and my mathematics background left me unable to let a claim stand unproved. So I proved them: with zero fusion, plasma, or nuclear-engineering training, I built [OpenFusion](#)  in 29 days (*inspect the evidence*)—a parametric kernel deriving the full tokamak plant, geometry through sensors, from a single design vector, with a browser 3-D instrument on top. And because speed without verification is noise, it audits itself: self-falsifying checks proven able to fail, and my own overclaims caught and retracted inside the record.
- **Now** — My work spans causal misalignment, mechanistic theory, public-input evaluation, and capability threat modeling. Having crossed rural central China, industrial Shenzhen, France, the United States, and sharp class and cognitive boundaries—and read deeply in history, philosophy, and critical theory—I understand the complexity of the real world: people inhabit different institutions, incentives, histories, values, and cognitive worlds, and technical systems must treat those differences as part of the problem rather than noise.

What I bring is not a narrow specialty or a fixed inventory of credentials. I am a frontier problem owner: I learn fast, find the structure, direct AI-agent execution, and hold the result to a standard my own artifacts enforce—claims that cannot survive their own falsification tests do not ship. My repositories carry my retractions alongside my results, because an artifact that has never been shown able to fail proves nothing. My value increases when the problem is important, ambiguous, cross-domain, and not yet decomposed.

My life was impossible. I made it possible. The problems this field faces are difficult. But difficulty presupposes possibility—and possibility permits hope. I regard every difficult-but-solvable problem as a privilege: however much it demands, there is a way through, and I know how to find or build it—as I always have.

Sincerely,
Ivan GAN